Satellite Collision Avoidance in Cislunar Space

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I. Introduction

The study of satellite collision avoidance has gained significance in recent decades, with the number of objects in Low-Earth Orbit (LEO) alone exceeding 25,000 according to the European Space Agency [1, 2]. While estimating the probability of collision (P_c) is a well-explored problem in near-Earth environments, with techniques dating back to Foster and Estes (1992) and further developed by researchers such as Alfano, Hall, Baars, and Casali, conjunction assessment in cislunar space and beyond remains a challenge [1, 3–5]. Cislunar space, which has significantly more complex gravitational dynamics than those seen in the near-Earth environment, provides a challenging new setting for testing optimal-policy-determination algorithms factoring in both the probability of a collision (as the existence of highly non-linear dynamics complicates the propagation of probability distributions), the cost (in propellant) of executing a maneuver, and the cost associated with deviating from a pre-determined mission orbit [6–8]. In particular, the design of algorithms that address scenarios involving an unresponsive object along the primary's trajectory will promote safety and limit disaster. We propose a recommended approach via extraction of an optimal or near-optimal policy from a Markov Decision Process designed to minimize probability of collision, costly maneuvers, and trajectory deviation.

II. Background and Related Work

The problem of conjunction assessment is a central tenet of the field of Space Situational Awareness (SSA). A *conjunction* is a close-approach event between two satellites (or, more generally, between two objects in space) [1]. The risk of an actual collision happening during these conjunctions is traditionally quantified using the probability of collision, or P_c, metric [1]. P_c is, in itself, a fairly difficult metric to calculate, with methods used depending on the conjunction geometry and perceived importance of the conjunction [4]. The most basic $2D-P_c$ method, developed by Foster and Estes in 1992, invokes the typical assumption that conjunctions happen over so short a time scale that the relative velocity between the two objects involved is constant; this allows marginalization of the joint position probability distribution along the direction of the relative velocity vector and simplifies conjunction assessment to an analytical integral over the hard-body radius (a collision sphere with a radius equal to the sum of the two satellite maximum radii) [9]. The constant-relative-velocity assumption applies to the vast majority (about 95%) of LEO conjunctions assessed by NASA's Conjunction Assessment Risk Analysis (CARA) group, with the remaining small fraction requiring more advanced semi-analytical and numerical methods [4]. Unfortunately, the dynamics of cislunar space make the task of conjunction assessment substantially more difficult [6]. The dynamics of cislunar space are dominated by the mutual effects of the Earth and the Moon, leading to trajectories very unlike those seen close to the Earth [7]. While a linear approach may still be valid for times very close to the time of close approach t_{CA} with spacecraft on significantly different trajectories, spacecraft in proximity for an extend duration of time (*e.g.*, two spacecraft on adjacent orbits in a given family) will violate this short-time assumption [7]. This necessitates developing new ways of estimating P_c, as well as of determining an optimal procedure for mitigating collision risks.

Most work to date on conjunction assessment has been focused on near-Earth satellites, in particular those in LEO [1]. The field of cislunar conjunction assessment is a developing one, with current approaches lacking in terms of orbit determination, state estimation, object catalogs, uncertainty propagation, and decision making [6]. This paper focuses on the uncertainty propagation and decision making aspects of this problem.

III. Problem Formulation

The problem of collision avoidance can be readily phrased as a Markov Decision Process (MDP). Here, it is assumed that one of the objects (the "primary") can be controlled by an owner/operator (O/O), while the other (the "secondary") either cannot or will not be controlled. It is additionally assumed that the primary can thrust along three orthogonal axes, $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$, and that these are used to drive trajectory changes. These thrusts define the action space, where there are 6 directions and 4 different levels of thrust magnitude for a total of 25 actions (including the zero-thrust action). The reward stems from three components: collision, maneuver cost, and trajectory deviation. The primary receives a large negative reward in a collision event, a small negative reward for executing a maneuver, a moderate negative award (proportional to the deviation) for deviating from the original planned trajectory, and zero reward otherwise. The state space is $\mathbb{R}^6 \times \mathbb{M}_6(\mathbb{R}) \times \mathbb{R}^6 \times \mathbb{M}_6(\mathbb{R}) \times \mathbb{R}^6 \times \mathbb{M}_6(\mathbb{R})$, where individual states $(\boldsymbol{\mu}_i, \Sigma_i)$ consist of the means and covariances of both objects; the third object is a duplicate of the first that does not carry out any actions and is used to track the ideal (original) primary trajectory at each point. Note that the state space is continuous and high-dimensional. The transition function is defined by the gravitational dynamics of cislunar space and maneuvers. It is implemented with a numerical integration scheme over CR3BP dynamics in cislunar space, which are described in detail in Section IV. We implement reward discounting with a value of 0.99 and discretization over small time steps (around 1 minute) to facilitate our computational calculations. The reward function itself is fairly complicated and is structured as follows:

$$R(s, a, s') = R_t(s) + R_c(a) + R_s(s, s') + R_w(s, s') + R_o(a, s'),$$

where R_t is the tracking reward, R_c is the control reward, R_s is the reward due to reward shaping, R_w is the stepwise tracking reward, and R_o is the on-track incentive reward. The tracking reward is computed as

$$R_t(s) = -w_r \|\Delta \mathbf{r}(s)\|^2 - w_v \|\Delta \mathbf{v}(s)\|^2,$$

where w_r and w_v are positive weights, $\Delta \mathbf{r}$ is the distance between the primary's position and ideal (*i.e.*, original trajectory) position at the next step, and $\Delta \mathbf{v}$ is the difference between the primary's velocity and ideal velocity at the next step. The control reward is given by

$$R_c(a) = -w_c \|a\|^2,$$

where w_c is a positive weight and a is the burn (action) ΔV. Potential reward shaping is used here with

$$R_s(s, s') = \gamma \Phi(s') - \Phi(s),$$

where

$$\Phi(s) = -k \left(\|\Delta \mathbf{r}(s)\|^2 + c \|\Delta \mathbf{v}(s)\|^2 \right);$$

k, c, and γ are all positive weights. The stepwise tracking reward incentivizes moving towards the original trajectory and is given by

$$R_w(s, s') = -k_p \left(\|\Delta \mathbf{r}(s)\| - \|\Delta \mathbf{r}(s')\| \right).$$

Finally, the on-track incentive reward prioritizes staying on the original trajectory once it has been returned to and is given by

$$R_o(a, s') = \begin{cases} w_o, & \|\Delta \mathbf{r}(s')\| < d \text{ and } a = \mathbf{0} \\ 0, & \text{otherwise} \end{cases},$$

where w_o and d are positive weights. Note the number of hyperparameters that must be tuned in this configuration!

IV. Solution Approach

Due to the complexity of cislunar space dynamics, we opted to run our simulations in python to utilize our modeling skills when writing code governing the nonlinear dynamics in these regions.

While initially we hoped that finding an optimal policy via either policy or value iteration would be sufficient to solve the deterministic version of this problem, we quickly realized that the dimensionality of the state space inhibits any enumeration over possible states, even with significant discretization; this motivated a search for alternative options. We made some attempt at discretizing at a coarse level and reducing to two dimensions, but were still struggling with staying true to the system's dynamics.

Our concern primarily lies with a set of very specific states that might potentially result in a collision. Value or policy iteration enumerates over states we do not care about; for example, states where objects are far away and moving aware from each other are not of interest. Thus, given the high spatial complexity of the problem and our interest in a subset of states, we naturally shift to online planning methods.

The primary tool of choice was Monte Carlo Tree Search (MCTS), which we fine-tuned for our problem. MCTS is capable of managing large state spaces, especially when the size of the action space is relatively small. Thus, it can adequately handle our continuous state space while still producing policy decisions in a reasonable period of time.

One consequence of the MCTS algorithm is that at any newly created node, a rollout policy chooses random actions. However, in this problem, the majority of actions incur a negative reward and make little to no sense. Thus, we introduced two heuristic rollout policies, one for the deterministic state case and one for the probabilistic state case. For the deterministic state case, 5% of the time a random action is taken, and the other 95% of the time, an action that balances between returning to the ideal trajectory and avoiding collision is chosen (weighted towards returning to the ideal trajectory to avoid incurring large negative penalties over time). For the probabilistic state case, 20% of the time a random action is chosen, and the other 80% of the time, an action that aligns with the current direction is chosen. This either encourages exploring states where the primary and secondary are close, thus resulting in a better reward, or facilitates actions that are more likely to keep the satellite closer to the original orbit.

Next, we used progressive widening to construct a more manageable tree. Ultimately, this was done to limit wasted time on the same node so that more future states could be explored with the same computational expense.

These updates to MCTS allowed us to use nonliear CR3BP dynamics, and as a result, we found little need to pursue subpar linear approximations, which do not hold universally in cislunar space. To implement these, we use the non-dimensional CR3BP equations of motion [7],

$$\begin{aligned}\ddot{x} - 2\dot{y} &= \frac{\partial U}{\partial x}\\ \ddot{y} + 2\dot{x} &= \frac{\partial U}{\partial y}\\ \ddot{z} &= \frac{\partial U}{\partial z},\end{aligned}$$

where

$$\begin{aligned}U(x,y,z) = \frac{1}{2}\left(x^2+y^2\right) + \\ \frac{1-\mu}{\sqrt{(x+\mu)^2+y^2+z^2}} + \\ \frac{\mu}{\sqrt{(x-1+\mu)^2+y^2+z^2}}.\end{aligned}$$

We integrate these numerically using an adaptive fourth/fifth-order Runge-Kutta scheme. Later, we introduce lower order approximations to save on computational time, but most of the deterministic initial approach is done with a more powerful fourth/fifth-order integrator.

We begin by approaching the deterministic case, where the state space contains only the mean vectors, namely, the position and velocity of the primary, secondary, and primary with no thrust. In this case, we experimented with vanilla MCTS before adding the modifications discussed earlier in this section.

Next, we add in an uncertainty element. In real-world scenarios, recorded measurements of a satellite's position from Earth will contain uncertainty, which is assumed to be Gaussian. Thus, we expand the state space to contain a covariance matrix associated with the mean vectors for position and velocity. This introduces complexity into the transition and reward functions.

With the added uncertainty, we now have an additional element that must be propagated through the CR3BP equations. However, the propagation of Gaussian covariance matrix through a nonlinear dynamical system does not preserve the linear relationship that a covariance matrix inherently expresses. Thus, we introduce an Extended Kalman Filter (EKF) to linearize the system around a Taylor expansion. Let $\mathbf{f} : \mathbb{R}^{12} \to \mathbb{R}^{12}$ be the function describing propagation of the mean vector integrated over timestep Δt, let $\tilde{\boldsymbol{\mu}}$ represent the result of changing the velocity vector $\boldsymbol{\mu}$, and let $\boldsymbol{\mu}^* = \mathbf{f}(\tilde{\boldsymbol{\mu}})$. From timestep k to timestep $k+1$, a covariance matrix Σ_k is propagated as

$$\Sigma_{k+1} = \Phi_k \Sigma_k \Phi_k^T + Q_k, \tag{1}$$

where

$$\Phi_t = Df(\boldsymbol{\mu}) = \left.\frac{\partial \mathbf{f}}{\partial \mathbf{x}}\right|_{\mathbf{x}=\boldsymbol{\mu}}$$

is the Jacobian matrix of the CR3BP dynamical system at the time associated with timestep k and

$$Q = \operatorname{diag}\left(q_x, q_y, q_z, q_{\dot{x}}, q_{\dot{y}}, q_{\dot{z}}\right)$$

is a diagonal process noise matrix consisting of uncertainties added to the propagation at each time step. Note that these are tuning parameters of this algorithm.

The reward function also changes to incorporate the rewards associated with the uncertain state. To do this, we define a vector to describe the relative position

$$\boldsymbol{\rho} \sim \mathcal{N}(\boldsymbol{\mu}_\rho, \Sigma_\rho), \tag{2}$$

where

$$\begin{aligned}\boldsymbol{\mu}_\rho &= \boldsymbol{\mu^*}_{\text{primary}} - \boldsymbol{\mu^*}_{\text{secondary}}\\ \Sigma_\rho &= \Sigma_{\text{primary}} + \Sigma_{\text{secondary}}.\end{aligned}$$

Let r represent the hard body radius, then, we can express the probability of collision, P_c, as $P(\|\rho\| < r)$. Note that

$$P(\|\boldsymbol{\rho}\| < r) = \int_{\|\mathbf{x}\|<r} \mathcal{N}(\mathbf{x}\,;\,\boldsymbol{\mu_\rho}, \Sigma_\rho)\,\mathrm{d}\mathbf{x}. \tag{3}$$

Methods to compute this integral do exist, however, they are complex and expensive, so we instead resort to approximations generated from the Mahalanobis metric. We project the problem onto the direction of the separation distance

$$\hat{\boldsymbol{\mu}}_\rho = \frac{\boldsymbol{\mu}_\rho}{\|\boldsymbol{\mu}_\rho\|}$$

and compute the squared Mahalanobis distance

$$D_M^2 = (\|\boldsymbol{\rho}\| < r)^2 \cdot \hat{\boldsymbol{\mu}}_\rho^T \Sigma^{-1} \hat{\boldsymbol{\mu}}_\rho \tag{4}$$

This computes the number of standard deviations the mean vector is from the hard body radius. To arrive at collision probability, we use a chi-squared distribution with 3 degrees of freedom (to approximate the 3 degrees of freedom in the state). We can then approximate P_c as

$$P_c \approx 1 - F_{\chi_3^2}(D_M^2). \tag{5}$$

V. Results

We tested our algorithm in the neighborhood of the L_4 equilibrium point since we can easily construct an orbit with stable behavior. We positioned the primary and secondary several kilometers apart with initial velocities pointing towards each other. We structured the problem such that roughly 5 time steps would result in a collision. This was done so that we would use less computational time exploring standard parts of the orbit, and the problem was designed with the vision of ease of generalization. The initial conditions can be applied to any scenario involving an orbit and a secondary object interfering.

Initially, we implemented a plain version of MCTS with a reward function that only depends on the maneuvers and the collision. At first, we did not consider the proximity of the primary to its original orbit. Doing so resulted in a policy that consistently performed well, but avoided the secondary by what was often a large margin. The results produced were highly dependent on initial parameters and the relative values of the various events in the reward function. Several examples of the results obtained are shown in Figures 1 and 2.

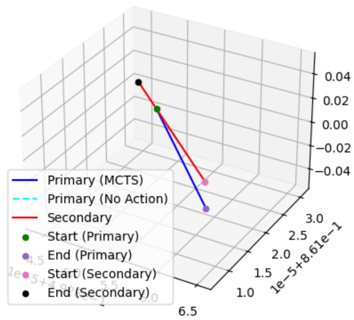

Fig. 1. Example trajectory given high cost associated with thrusting actions relative to the penalty associated with collisions. The units here are nondimensional CR3BP distances from the origin.

When the cost of executing a maneuver is large, we often see the trajectory include one initial action. This also requires significant simulation time that allows MCTS to explore deep enough to choose an action that avoids a collision several steps in advance.

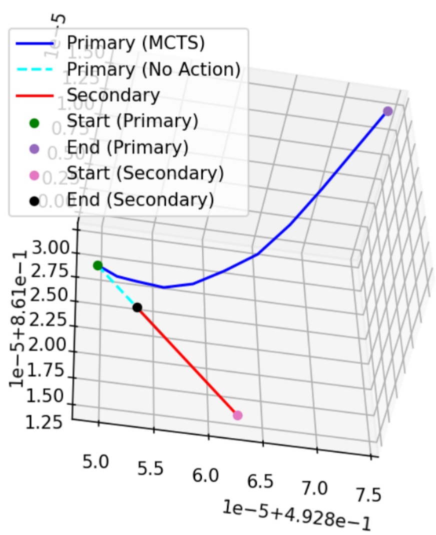

Fig. 2. Example trajectory with large negative rewards associated with a collision. Again, these units are nondimensional CR3BP distances from the origin.

When the relative cost of a collision is large, this dominates, and we see a tendency for the object to deviate very far from the original course. Often, the object continues to thrust even after it appears to have safely escaped danger of collision. This might be a good policy if the secondary is adversarial in nature, or in a rare case when the secondary is acting with some form of pursuit. However, this is not a common phenomenon, and is practically nonexistent in the current state of cislunar space practices. While this behavior is logical in some far-fetched situations, in non-adversarial instances, this divergence from the orbit is problematic, as it derails the satellite from its original track, causing difficulty in accomplishing the initial purpose or task. This illustrates the need for proper reward structuring and fine tuning.

When we introduced more complexity into the MCTS function, we observed a better result. The introduction of the heuristic rollout policy allowed for more accurate approximations after each new node is created, which gave us better far-out rewards. Next, the use of proximity widening helped to combat the extra computational expense that arises from a more complex rollout. These modifications resulted in a slightly better policy, which is shown in Fig. 3.

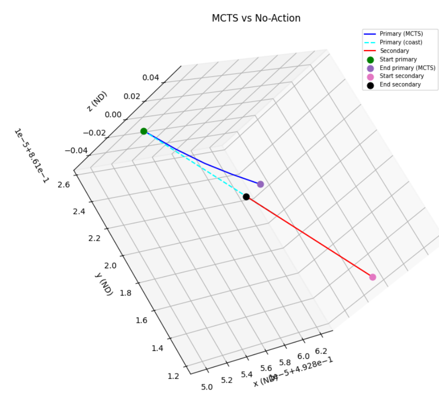

Fig. 3. Adding a heuristic rollout and progressive widening resulted in a slight improvement. Notably, these modifications do not solve the problem of tuning the reward weights; small action penalties or large collision penalties will impact the trajectory depicted.

In both policies, we realized that a critical flaw in our reward system lies in the incentive for a satellite to significantly deviate from orbit. While it is desirable to avoid a crash, it is not beneficial to veer drastically off course. Thus, we added another component to our reward function: a distance penalty, as described in Section III. This encourages the satellite to return towards the original orbit while still allowing for a large penalization of a collision. The results of a typical run with this more advanced deterministic state policy are shown in Figure 4.

After we implemented a deterministic approach, we used our most polished version of MCTS to carry out a probabilistic interpretation. Note that some simplifications were originally tried in case there were any improvements arising from discrepancies between the cases, but none were found, and thus, the results presented involve MCTS with a heuristic rollout and progressive widening. Without any penalization related to the original orbit, we observed expected and well-controlled behavior, as displayed in Fig. 5.

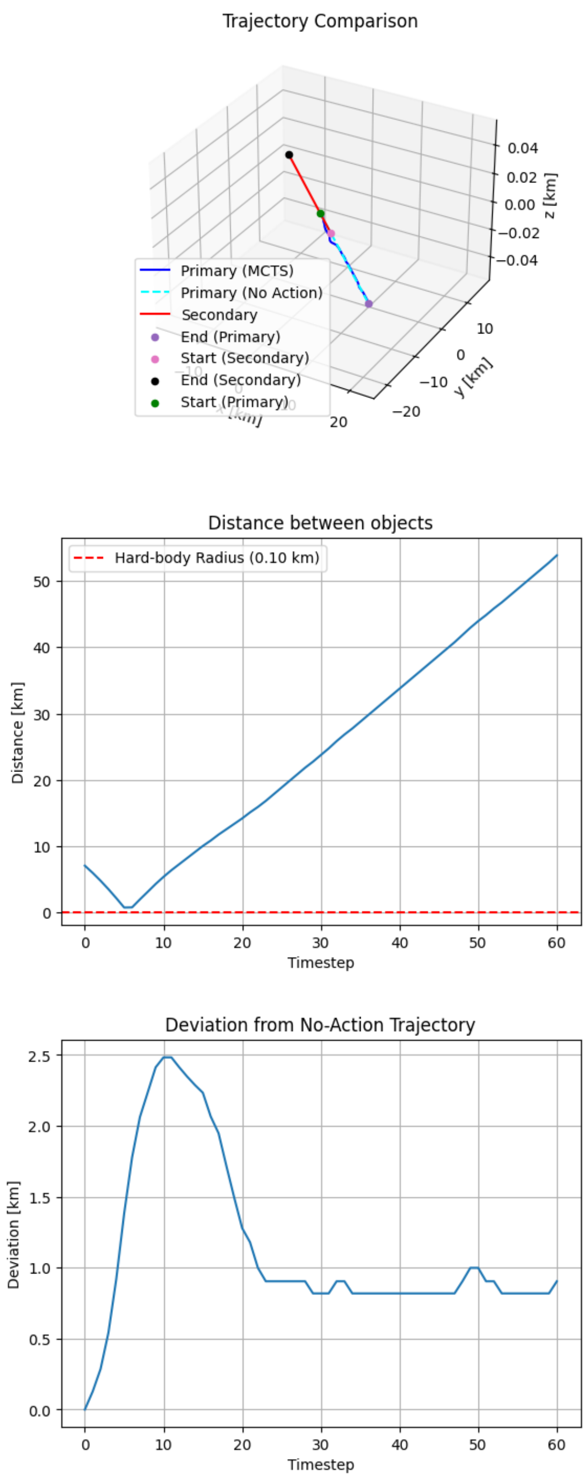

Fig. 4. A typical run of the advanced deterministic state policy. The top plot illustrates how, after a short deviation (near the beginning) to avoid a collision, the primary tracks back to the original trajectory. The bottom plot illustrates this in more detail, showing the steady-state error resulting from the policy balancing thrust and drift penalties after a maneuver. The middle plot verifies that the objects never collided.

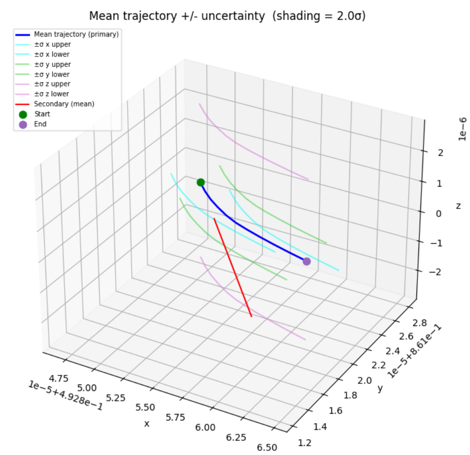

Fig. 5. The probabilistic simulation exhibits similar behavior with additional caution. Uncertainties are plotted alongside the trajectory.

While the confidence bounds demonstrate the avoidance of the high probabilities of collision, we also measure the distance from the primary to the hard body radius over time. When combined with the uncertainty thresholds, we observed that the primary avoided any negative reward associated with regions of higher P_c; this phenomenon is shown in Fig. 6. This is ideal behavior and is also customizable depending on the weight associated with the penalty for for P_c.

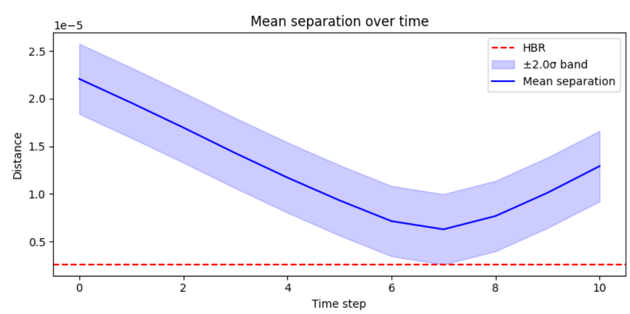

Fig. 6. Distance from a collision over time. Distance is measured in terms of nondimensional CR3BP units, and thus maintains a small scale.

Observe that the primary takes action and avoids negative rewards when the uncertainty threshold encounters the hard body radius. From here, it continues along, and as the secondary retreats, the distance increases, and thus, the threat of collision grows distant.

The effect of the uncertainty makes the probabilistic approach less susceptible to parameters that determine the relative importance of the components in the reward function. If these are adjusted such that a collision is deemed much more dangerous, the trajectory is relatively similar, as illustrated in Fig. 7.

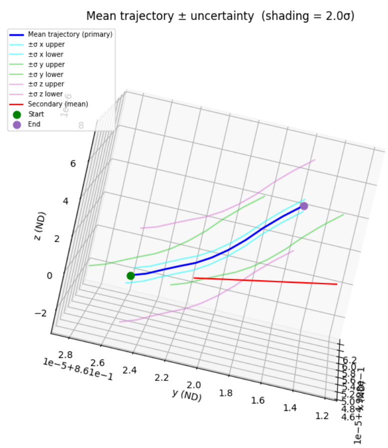

Fig. 7. Even with more emphasis placed on collision avoidance, the trajectory still evolves in a more stable manner.

VI. Conclusions and Future Work

We investigate various online planning approaches for constructing a policy when navigating collision concerns surrounding unresponsive secondary obstacles in an orbital path. We begin with plain MCTS and a deterministic worldview and modify it with heuristic policies and progressive widening to improve computational efficiency. We introduce a reward correcting mechanism to balance the need between staying close to the original orbit and avoiding collision. We transition to a probabilistic interpretation, where we demonstrate more observed stability and more caution around regions of uncertainty. Ultimately, we produce a framework capable of finding solid policy options for addressing a risky collision situation.

In the project proposal, we were more optimistic about the utility of value and policy iteration; we realized that this was intractable and pivoted to solely online planning. MCTS was the main choice, and we tested different implementations and extensions of it. For the probabilistic interpretation, we altered the plan to estimate P_c, and relied upon a slightly different method than 2D-P_c. However, we still projected onto the relative position vector; the difference arises in our final probability estimate, which we derive from the c.d.f. of a chi-squared distribution. We also deviate with our restructuring of the reward function, which we introduce to mitigate potential issues that arose once we observed the results of the simulations. We were able to exceed the expectations of the original proposal by discarding the need for a linearized model, using fully nonlinear dynamics instead, and exploring the deterministic case and probabilistic case both more thoroughly than originally planned, although at the cost of focusing on MCTS.

However, there are many directions that could potentially improve the robustness, efficiency, and usability of this methodology. Future work could improve upon the the parameter tuning, integration of new and improved reward structures, and, most notably, addressing the issue of covariance matrix propagation. Propagation of the covariance matrices associated with the Gaussian distributed state results in accuracy loss over longer periods of time. Since nonlinear dynamics ultimately erode the reliability of these dynamics, we believe valuable future work lies in attempts to re-cast the dynamics of the CR3BP in terms of local orbital elements (LOEs), hopefully allowing for longer-term propagations of orbital states mirroring the methods used for LEO conjunction assessment [1, 10]. A comparison of these states would illustrate the utility of using these sets for analysis. Additionally, this work focuses solely on one aspect of cislunar conjunction assessment; not included is an analysis of covariance realism or effects from missing data. A more robust algorithm would incorporate the effects of imperfect data, likely using methods in state estimation and orbit determination theory to do so [11, 12].

VII. Contributions and Release

Erick was the primary researcher for the project; he worked on the problem setup, initial examples for the specific L_4 simulation, reward function adaptations, and code organization. Elizabeth produced code defining MCTS, plotting schemes, MCTS modifications, and the structure and implementation for the probabilistic interpretation. Erick focused on the deterministic case, while Elizabeth focused on the probabilistic case. Both contributed equally to both the implementation and the writing of this report.

The authors grant permission for this report to be posted publicly. Repository code is available here.

All algorithms and code used in this project was implemented from scratch by both team members. Existing code from Erick's previous classes and research was used to implement the numerical integrators used to propagate the CR3BP dynamics, but aside from this, all code was handwritten in python. Copilot was used for debugging reward structure issues, but otherwise, no AI was used in the creation of this project or report.

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